

RAMCO INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Academic Year: 2019 - 2020 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. Computer Science and Engineering

Course Code & Title: CS8451 & Design and Analysis of Algorithms

Name of the Faculty member: Ms.P.Jothi Thilaga, AP/CSE

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Innovative Practice: Student Team Achievement Divisions (STAD)

Topic: Dynamic Programming and Greedy Technique

Type of Learning:

Collaborative Learning

Learning Objectives:

O1: To make the students to solve the problems using Dynamic programming and Greedy Technique.

Description:

STAD is considered as one of the most researched, simplest and straightforward collaborative learning. The students are placed in small groups. The class in its entirety is presented with a lesson and students are subsequently tested. Individuals are graded on the team's performance. Although the tests are taken individually, students are encouraged to work together to improve the overall performance of the group. It is basically a team work, but students are graded individually according to their contribution that they make towards their team.

Justification for chosen the topic:

Dynamic Programming and Greedy Technique are the most important techniques in design and analysis of algorithms. This activity makes the students to get a sound knowledge to solve the problems using these techniques. Students think individually and solve the problems by discussing with their team members can improve their knowledge in the particular topic.

Implementation of STAD:

- Initially the groups were formed based on the student's willingness. Eleven teams were formed with a group of 4 or 5 students. Team members selected a team name based on their interest.

- Each team was assigned with a different set of questions. They have to solve the given set of questions by discussing with their team members. 20 minutes were allocated to solve the given set of question.
- Finally each team discusses the methods and steps to solve the allotted problems within the given time to the entire class.
- Based the rubrics designed, each teams were graded. The rubric for this activity is given in the Table 1.1.

Table 1.1: Rubric for STAD Activity

Evaluation criteria	Excellent 5.0 to 4.0 pts	Good 3.0 to 2.0 pts	Satisfaction 2.0 to 0 pts	Marks
Knowledge in the given topic	Every Student in the team can able to solve the problems individually.	Student in the team can able to solve the problems individually, but needs some additional support.	Student in the team feel some difficulties in solving the problems individually.	5
Time Management	Answer all the questions correctly within the allotted time	Answer the questions partially within the allotted time	Answer the questions beyond the allotted time	5
Group Coordination	Each individual performed excellently and coordinated well with the team members	Each individual performed good and coordinated well with the team members	Each individual performed good and coordination among the team members is lacking	5
Group Presentation	The presentation was divided equally to individual in a group. The individual performed excellently.	The presentation was somewhat divided to an individual in a group. An individual was somewhat performed well	The presentation was not equally divided into an individual in a group. An individual was not performed well	5
Total				20

The individual score will be calculated based on the above rubrics such as Knowledge in the given topic, Time Management, Group Coordination, Group Presentation. The final team score was calculated by taking the average of the individual scores in each team and it is listed in the Table 1.2.

Table 1.2: Team wise Scores

Sl. No.	Reg. No.	Name of the Student	Group Name	Marks (20)
1.	953618104004	ARAVINTH R	Master	18
2.	953618104031	SANTHOSH G		
3.	953618104036	SIVABALAN M		
4.	953618104022	NANNEBOINA SUMANTH		
5.	953618104034	SHIVANESH S	Valimai	15
6.	953618104035	SIVA T		
7.	953618104041	SRIRAM R		
8.	953618104005	ARUNKUMAR P		
9.	953618104001	ABDUL MALIK S	Chill Bros	16
10.	953618104002	ABISHEK S		
11.	953618104021	NAMBIRAJAN A		
12.	953618104043	SURIYAA M		
13.	953618104008	ESAKKIRAJA A	Phoenix	18
14.	953618104014	KANNAN B		
15.	953618104017	MATHEESH G		
16.	953618104026	RAM BABU B		
17.	953618104037	SIVASANKAR M	Four Tigers	17
18.	953618104027	RANJITHKUMAR M		
19.	953618104042	SRI SHANKAR R		
20.	953618104018	MOHAMED YAASEEN Z		
21.	953618104046	UMASUDAN G	Spartans	16
22.	953618104019	MOHAN BABU S		
23.	953618104020	MOHAN KUMAR D		
24.	953618104044	SURYAKUMAR S		
25.	953618104010	GNANAMOORTHY VIDHIA	Hifi	19
26.	953618104012	HARSHINI A S		
27.	953618104039	SOWMIYA S		
28.	953618104045	UMASRI R		
29.	953618104048	VINDHIYA SHREE K		
30.	953618104047	VAISHNAVI A	Sparkle	20
31.	953618104015	KARTHIGA M		
32.	953618104049	VISHNU PRABA V		
33.	953618104701	VIJAYALAKSHMI S		
34.	953618104038	SNEKA S		

35.	953618104009	EVANGELIN BLESSY A	Star Wary	18
36.	953618104007	DHARSNI R		
37.	953618104006	ASWINI B		
38.	953618104003	ANUGRAGHA A		
39.	953618104040	SRI DARSHINI M B		
40.	953618104013	HEMALATHA S	V5 Rockers	20
41.	953618104024	PADMASUGI R		
42.	953618104029	SABNA DEVI P		
43.	953618104016	KAVIMALAR S		
44.	953618104028	RITHIKA M		
45.	953618104033	SHANTHI PRIYA P	Squadron	18
46.	953618104025	RAHINI G		
47.	953618104023	NIVETHITHA R		
48.	953618104030	SAMYUKTHA VIKASHINI S		
49.	953618104032	SHALINI P		

Glimpse of STAD Activity



Outcomes:

- At the end this activity each students in the team can be able to solve the assigned problems to their team and know the concepts clearly and to able explain the same to the other students.
- This activity makes the students to understand the way to solve the problems using dynamic programming and greedy technique.

Relevance to POs

Objective	PO1	PO2	PO3	PO4	PO5	PO9	PO10
O1	3	3	2	2	2	3	2

Reflective Report:**Identified Problems**

- Some of the students were not aware about the STAD activity. It takes additionally few minutes to explain them about this activity.
- Presentation by the entire team takes more time since they deviate from key concepts and explaining all the concepts.

Initiatives to address the problems

- Addressed the students that the individual participation in the team can only make the team grades good.
- Make the students to know the impact and importance of sharing their views and understanding related to the topic and made them involve in the activity.

Post-implementation

- Students were actively participated in this activity.
- From this activity, the students can get more clarity in the particular topic by solving problems and discussing with the other students in the class.

References:

- <https://www.ritrjpm.ac.in/images/computer-science/CS8601-Mobile%20Computing-STAD.pdf>
- http://cw.routledge.com/textbooks/9780415965286/studentresources_chapter8_cooperative.asp
- <https://iopscience.iop.org/article/10.1088/1757-899X/336/1/012027/pdf>

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